

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech Winter 2025 - 26 Examination

Semester: 7
Subject Code: 203105493
Subject Name: Big Data on Cloud

Date: 19/12/2025
Time: 2:00pm to 4:30pm
Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1 is compulsory**, From Section B, **Q.1 is compulsory**.
3. Figures to the right indicate full marks.
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-63

SECTION-A		Marks	CO	BT
Q.1	Answer the following questions.			
	A. 1. What is Distributed Data Processing? 2. Define Data Warehouse. 3. What is Streaming Data Clustering?	[06]	1	1
	B. 1. List two Cloud Deployment Models with real life examples. 2. Define Big Data Modeling. 3. What is Cloud Infrastructure Management?	[06]	1	1
Q.2	A. What is Big Data? Explain the advantages and limitations of Big Data with suitable examples.	[04]	2	2
	B. Explain how data is pre-processed before analysis, with examples.	[05]	3	3
	OR			
	B. Explain Noise and Outlier Detection methods used in data mining.	[05]	3	4
Q.3	A. Explain Classification and Clustering in Big Data with suitable examples.	[04]	2	2
	B. Compare Supervised vs. Unsupervised Learning in detail with real-world applications.	[05]	3	4
	OR			
	B. Discuss the role of Collaborative Filtering in recommendation systems with an example.	[05]	3	3

SECTION-B		Marks	CO	BT
Q.1	Answer the following questions.			
	A. 1. Give two examples of a key-value NoSQL database. 2. What is the main advantage of in-memory computation in Spark? 3. Give two advantages of batch processing.	[06]	5,2	1
	B. 1. Why is data lineage important in Spark? 2. Name the categories of NoSQL databases. 3. Give two advantages of real-time processing.	[06]	2,5	1
Q.2	A. Explain in detail a real-world scenario where real-time processing is essential, and explain why batch processing would not be suitable.	[04]	2	3
	B. Compare RDBMS and NoSQL databases in terms of structure, scalability, and performance, with examples.	[05]	5	4
	OR			
	B. Explain the difference between Transformations and Actions in Spark with examples.	[05]	2	4
Q.3	A. Describe a scenario in healthcare where real-time processing is preferred over batch processing.	[04]	2	4
	B. Explain the concept of in-memory computation in detail and its advantages and disadvantages.	[05]	2	2
	OR			
	B. What is MongoDB, and how is it different from other NoSQL databases? List the key features of MongoDB and explain its advantages.	[05]	5	3